1. **Resolve Merge Conflicts**
   - Create a merge conflict intentionally (two users editing the same line).
   - Resolve the conflict and push the changes.

Step 1: First created 2 branches from main branch **git branch –c "name" (nova & mercury)**
Step 2: pushed the branches to github **(git -u push origin "branch name")**
Step 3: checkout to mercury branch and edited file → git add "file" → git commit -m "message"
Step 4: checkout to nova branch and edited file → git add "file" → git commit -m "message"
Step 5: stay in one of the branches and use merge command **(git merge nova)** and you will get the conflict message.
Step 6: edit the file using editor I used **vi filename** edited the file content as per my requirement then save file → **git add filename → git commit – m "conflict resolved"**
Step 7: git push

Results Below:

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git branch
* main


User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git branch -c nova

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git branch -c mercury

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git branch
* main
  mercury
  nova

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$
```

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git push -u origin nova
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'nova' on GitHub by visiting:
remote:      https://github.com/viquar44/October-23-newtask/pull/new/nova
remote:
To https://github.com/viquar44/October-23-newtask.git
 * [new branch]      nova -> nova
branch 'nova' set up to track 'origin/nova'.

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git push -u origin nova
branch 'nova' set up to track 'origin/nova'.
Everything up-to-date

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git push -u origin mercury
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'mercury' on GitHub by visiting:
remote:      https://github.com/viquar44/October-23-newtask/pull/new/mercury
remote:
To https://github.com/viquar44/October-23-newtask.git
 * [new branch]      mercury -> mercury
branch 'mercury' set up to track 'origin/mercury'.
```

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git checkout nova
Switched to branch 'nova'
Your branch is up to date with 'origin/nova'.

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (nova)
$ ls
README.md  code.txt  file1  file2  sample.txt

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (nova)
$ vi sample.txt

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (nova)
$ git checkout mercury
M       sample.txt
Switched to branch 'mercury'
Your branch is up to date with 'origin/mercury'.

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (mercury)
$ ls
README.md  code.txt  file1  file2  sample.txt

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (mercury)
$ vi sample.txt

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (mercury)
$
```

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (nova)
$ git merge mercury
Auto-merging sample.txt
CONFLICT (content): Merge conflict in sample.txt
Automatic merge failed; fix conflicts and then commit the result.

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (nova|MERGING)
$ git
```

UCRT64:/c/Users/User,

```
1 rose
2 lily
3 tulip
4 gerbera
<<<<<<< HEAD
5 eustoma
6 leaves
=======
5 cape mix
6 salix
>>>>>>> mercury
~
```

UCRT64:/c/Users/User/Desk

```
1 rose
2 lily
3 tulip
4 gerbera
5 eustoma
6 cape mix
~
~
```

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (nova|MERGING)
$ git status
On branch nova
Your branch is ahead of 'origin/nova' by 1 commit.
  (use "git push" to publish your local commits)

You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
        both modified:   sample.txt

no changes added to commit (use "git add" and/or "git commit -a")

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (nova|MERGING)
$ git add .

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (nova|MERGING)
$ git commit -m "Edited conflicts"
[nova 6335752] Edited conflicts

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (nova)
```

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (nova|MERGING)
$ git commit -m "Edited conflicts"
[nova 6335752] Edited conflicts

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (nova)
$ git push
Enumerating objects: 11, done.
Counting objects: 100% (11/11), done.
Delta compression using up to 8 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (9/9), 883 bytes | 441.00 KiB/s, done.
Total 9 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
To https://github.com/viquar44/October-23-newtask.git
   9877862..6335752  nova -> nova
```

2. **Recover Deleted Branch**
    - Delete a local branch and then recover it using the reflog.

Step 1: created a branch **git branch sun → git push origin -u sun**
Step 2: Deleted the branch **git branch -d sun**
Step 3: check the log id by using command a **git reflog**
Step 4: Recover the branch by using a command **git checkout -b sun "reflog id"**

Results below:

UCRT64:/c/Users/User/Desktop/Task-23/October-23-newtask

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git branch
* main
  mercury
  nova

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git branch sun

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git branch
* main
  mercury
  nova
  sun

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git push -u origin sun
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'sun' on GitHub by visiting:
remote:      https://github.com/viquar44/October-23-newtask/pull/new/sun
remote:
To https://github.com/viquar44/October-23-newtask.git
 * [new branch]      sun -> sun
branch 'sun' set up to track 'origin/sun'.

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$
```

UCRT64:/c/Users/User/Desktop/Task-23/October-23-newtask

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git branch -d sun
Deleted branch sun (was 9877862).

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git push origin -d sun
To https://github.com/viquar44/October-23-newtask.git
 - [deleted]         sun

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$
```

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git checkout -b sun 9877862
Switched to a new branch 'sun'

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (sun)
$ git branch
  main
  mercury
  nova
* sun

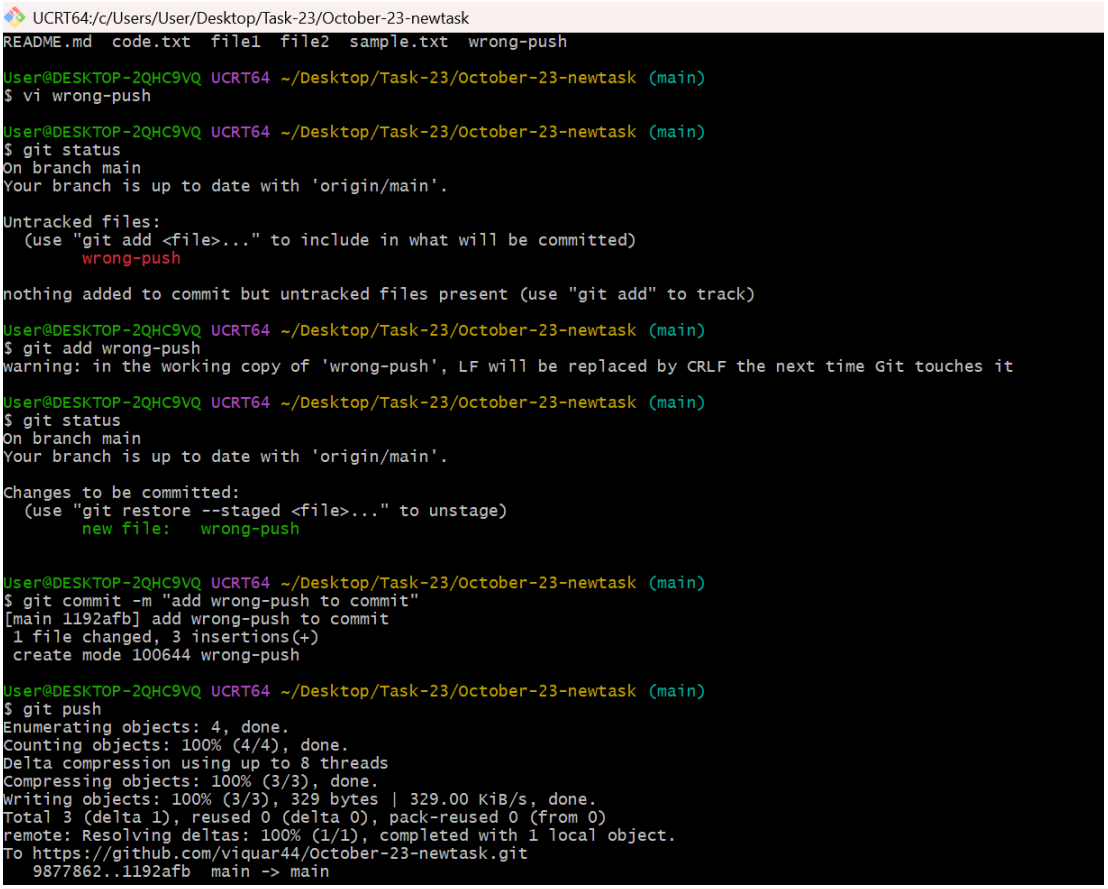
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (sun)
$
```

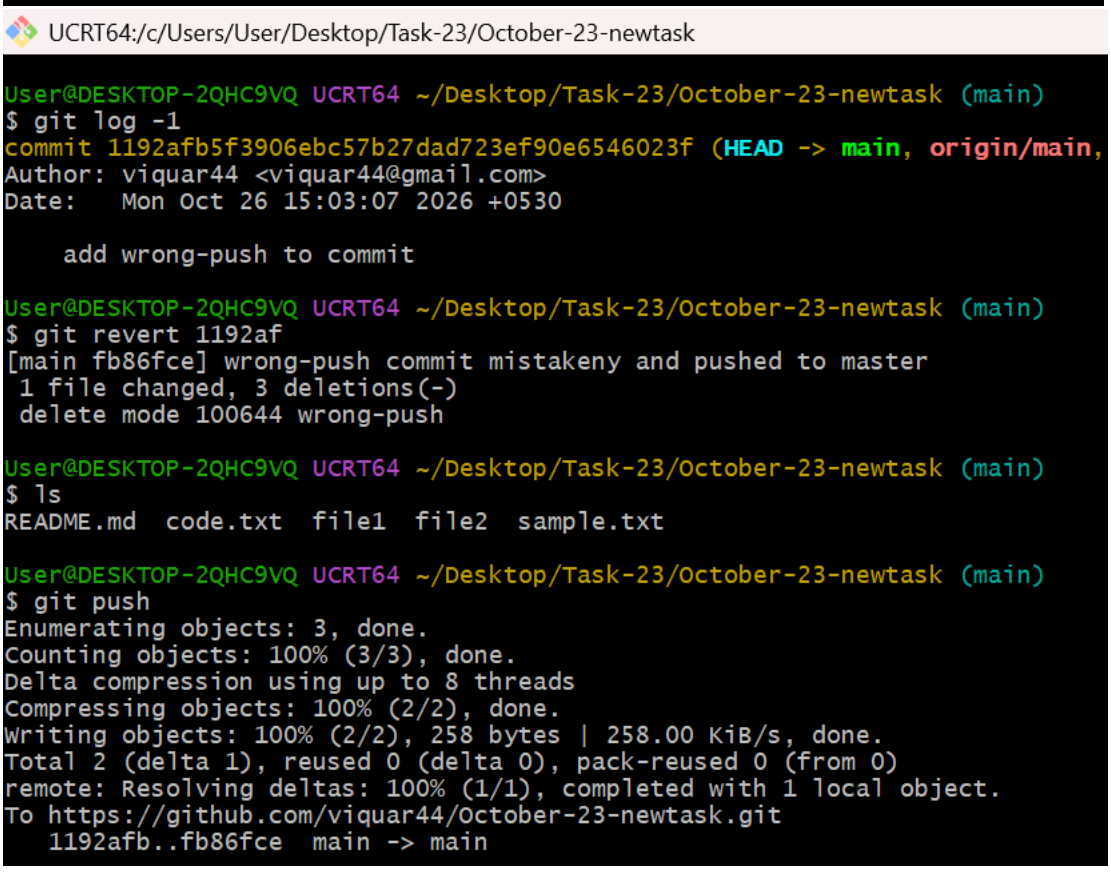
3. **Undo Wrong Push**
    - Push a wrong commit to GitHub, then undo it without losing history.

Step 1: Created a file → **vi wrong-push → git add . → git commit -m "messege" → git push**

Step 2: Used **git revert "commit id"** → edit the message → file deleted → **git push**

Results below:





4. **Amend a Commit**
   - Make a commit, then add a missing file to it using `git commit --amend`.

Step 1: created a file → **git add . → git commit -m "message"**

Step 2: created another file → **git add . → git commit - -amend**

So if you use git commit - -amend it will edit last commit editor will open just edit the message and save the file then check log **git log** you will see another file added to same commit.

Results below:

UCRT64:/c/Users/User/Desktop/Task-23/October-23-newtask

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ ls
README.md  amend  code.txt  file1  file2  sample.txt

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        amend

nothing added to commit but untracked files present (use "git add" to track)

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git add .

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git commit -m "amend added to commit"
[main 1f8b867] amend added to commit
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 amend

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ vi new-amend

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        new-amend

nothing added to commit but untracked files present (use "git add" to track)
```

UCRT64:/c/Users/User/Desktop/Task-23/October-23-newtask

```
amend and new-amend dded to commit

# Please enter the commit message for your changes. Lines starting
# with '#' will be ignored, and an empty message aborts the commit.
#
# Date:      Mon Oct 26 15:26:26 2026 +0530
#
# On branch main
# Your branch is ahead of 'origin/main' by 1 commit.
#   (use "git push" to publish your local commits)
#
# Changes to be committed:
#       new file:   amend
#       new file:   new-amend
#
~
~
~
```

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git add .
warning: in the working copy of 'new-amend', LF will be replaced by CRLF the next time Git touches it

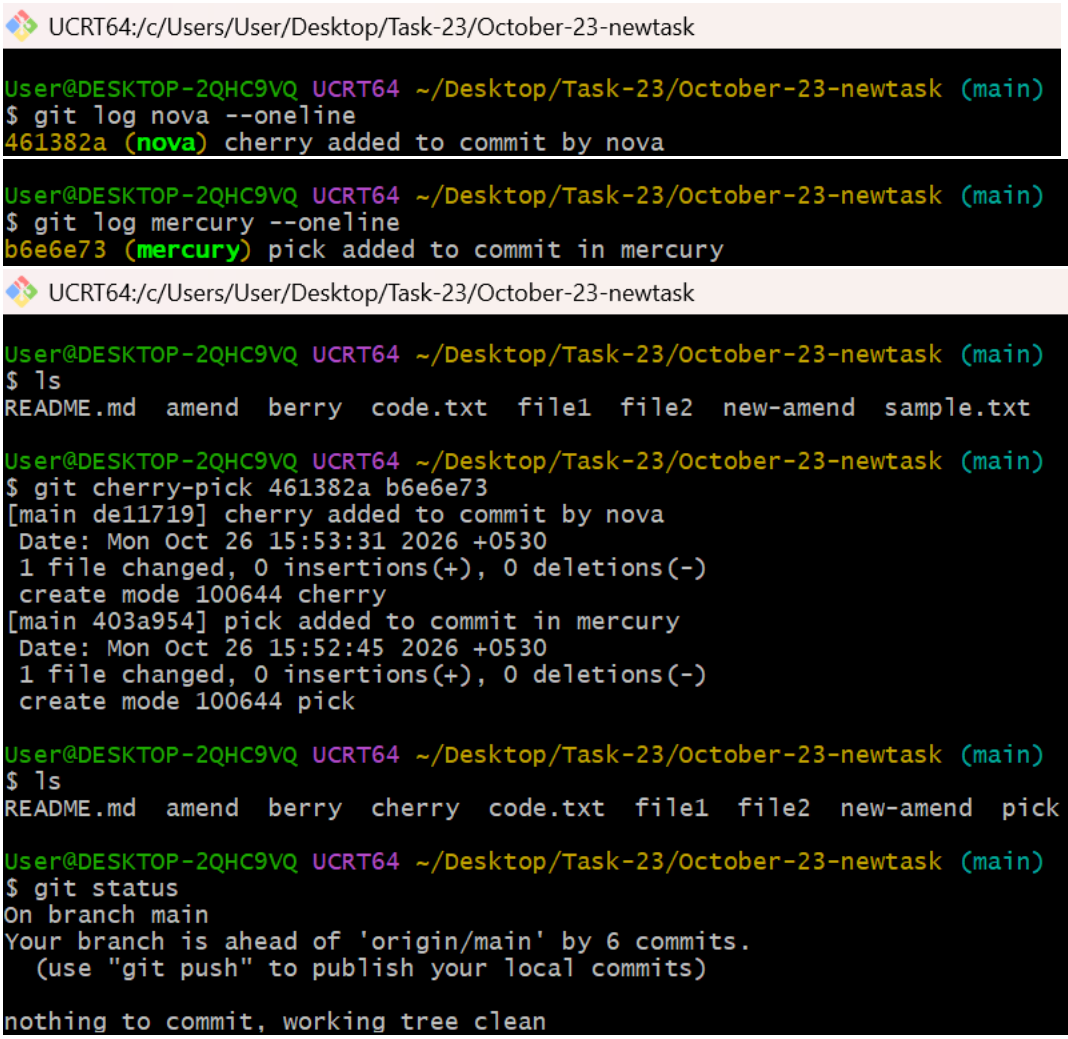
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git commit --amend
[main e85cd9a] amend and new-amend dded to commit
 Date: Mon Oct 26 15:26:26 2026 +0530
 2 files changed, 1 insertion(+)
 create mode 100644 amend
 create mode 100644 new-amend

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git log -oneline
fatal: unrecognized argument: -oneline

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git log --oneline
e85cd9a (HEAD -> main) amend and new-amend dded to commit
fb86fce (origin/main, origin/HEAD) wrong-push commit mistakeny and pushed to master
1192afb add wrong-push to commit
9877862 (origin/sun, sun) changes
```

5. **Cherry-pick a Commit**
    - Take a specific commit from one branch and apply it to another branch.

Step 1: check the log of other branches to pickup a file **[git log "branch name" –oneline]**
Step 2: copy the **commit ID "6 digit"**
Step 3: used a command **[git cherry-pick "commit id"]**
Step 4: check if that file came to your repo by using a **command ls**
In my case I have copied 2 files from 2 different branches each branch one so I have used Command as you can see in below picture [**git cherry-pick 461382a b6e6e73**]



6. **Interactive Rebase**
   - Reorder and squash multiple commits into a single clean commit.

Step 1: Get the logs **[git log -3]** , in my case I have choose only last three logs.

Step 2: I have made 1 commit instead of 3 different commits for 3 files, So I have used rebase command **[git rebase -i HEAD~3]** it will pull editor page and where you have to change **pick** to **squash** in 2nd and 3rd line, then you will open another page where you will edit or change commit message.

Step 3: Now check the last log **[git log -1]** so you will see single commit and changed commit message

Results below:

UCRT64:/c/Users/User/Desktop/Task-23/October-23-newtask

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git log
commit 403a95493ac7829315f2f078b43d4f1a79a7d0d6 (HEAD -> main, origin/main, origin/HEAD)
Author: viquar44 <viquar44@gmail.com>
Date:   Mon Oct 26 15:52:45 2026 +0530

    pick added to commit in mercury

commit de11719c4163d3449f573b3242e0a49a313d35bc
Author: viquar44 <viquar44@gmail.com>
Date:   Mon Oct 26 15:53:31 2026 +0530

    cherry added to commit by nova

commit 2c5d443967f60508b7b23c28d8c1ea6118d3ea4e
Author: viquar44 <viquar44@gmail.com>
Date:   Mon Oct 26 15:54:13 2026 +0530

    berry added to commit by main
```

UCRT64:/c/Users/User/Desktop/Task-23/October-23-newtask

```
pick 2c5d443 # berry added to commit by main
pick de11719 # cherry added to commit by nova
pick 403a954 # pick added to commit in mercury

# Rebase 913ed19..403a954 onto 913ed19 (3 commands)
#
# Commands:
# p, pick <commit> = use commit
# r, reword <commit> = use commit, but edit the commit message
# e, edit <commit> = use commit, but stop for amending
# s, squash <commit> = use commit, but meld into previous commit
# f, fixup [-C | -c] <commit> = like "squash" but keep only the previous
#                    commit's log message, unless -C is used, in which case
#                    keep only this commit's message; -c is same as -C but
#                    opens the editor
# x, exec <command> = run command (the rest of the line) using shell
# b, break = stop here (continue rebase later with 'git rebase --continue')
# d, drop <commit> = remove commit
# l, label <label> = label current HEAD with a name
# t, reset <label> = reset HEAD to a label
# m, merge [-C <commit> | -c <commit>] <label> [# <oneline>]
#         create a merge commit using the original merge commit's
#         message (or the oneline, if no original merge commit was
#         specified); use -c <commit> to reword the commit message
# u, update-ref <ref> = track a placeholder for the <ref> to be updated
#                       to this position in the new commits. The <ref> is
#                       updated at the end of the rebase
```

UCRT64:/c/Users/User/Desktop/Task-23/October-23-newtask

```
pick 2c5d443 # berry added to commit by main
squash de11719 # cherry added to commit by nova
squash 403a954 # pick added to commit in mercury

# Rebase 913ed19..403a954 onto 913ed19 (3 commands)
#
# Commands:
# p, pick <commit> = use commit
# r, reword <commit> = use commit, but edit the commit message
# e, edit <commit> = use commit, but stop for amending
# s, squash <commit> = use commit, but meld into previous commit
# f, fixup [-C | -c] <commit> = like "squash" but keep only the previous
#                    commit's log message, unless -C is used, in which case
#                    keep only this commit's message; -c is same as -C but
#                    opens the editor
```

UCRT64:/c/Users/User/Desktop/Task-23/October-23-newtask

```
# This is a combination of 3 commits.
# This is the 1st commit message:

berry added to commit by main

# This is the commit message #2:

cherry added to commit by nova

# This is the commit message #3:

pick added to commit in mercury

# Please enter the commit message for your changes. Lines starting
# with '#' will be ignored, and an empty message aborts the commit.
#
# Date:      Mon Oct 26 15:54:13 2026 +0530
#
# interactive rebase in progress; onto 913ed19
# Last commands done (3 commands done):
#    squash de11719 # cherry added to commit by nova
#    squash 403a954 # pick added to commit in mercury
# No commands remaining.
# You are currently rebasing branch 'main' on '913ed19'.
#
# Changes to be committed:
#       new file:   berry
#       new file:   cherry
#       new file:   pick
#
```

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git log
commit 46f489b2742563bb9d4f8a508b0c365722583816 (HEAD -> main)
Author: viquar44 <viquar44@gmail.com>
Date:   Mon Oct 26 15:54:13 2026 +0530

    berry , cherry & pick added to commit by main
```

7. **Tagging & Release**
   - Create a version tag (v1.0), push it to GitHub, then delete and restore it.

Step 1: check if you have existing task **[git tag]** ; in my case I don't have and created one **[git tag v1.0]**
Step 2: pushed the tag to remote **[git push origin v1.0]** ; checked in GitHub.
Step 3: deleted the tag using a command **[git tag -d v1.0]** → delete from **[git push origin -d v1.0]**
Step 4: Restore the code using ID **[git tag v1.0 a6f6947]** → then push to remote **[git push origin v1.0]**

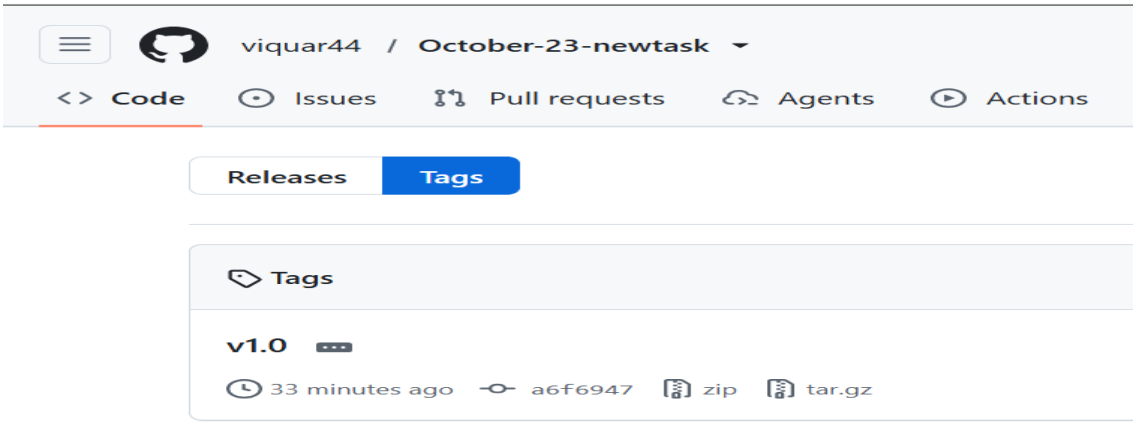
UCRT64:/c/Users/User/Desktop/Task-23/October-23-newtask

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git tag

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git tag v1.0

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git tag
v1.0


User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git push origin v1.0
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/viquar44/October-23-newtask.git
 * [new tag]         v1.0 -> v1.0
```




```
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git tag -d v1.0
Deleted tag 'v1.0' (was a6f6947)

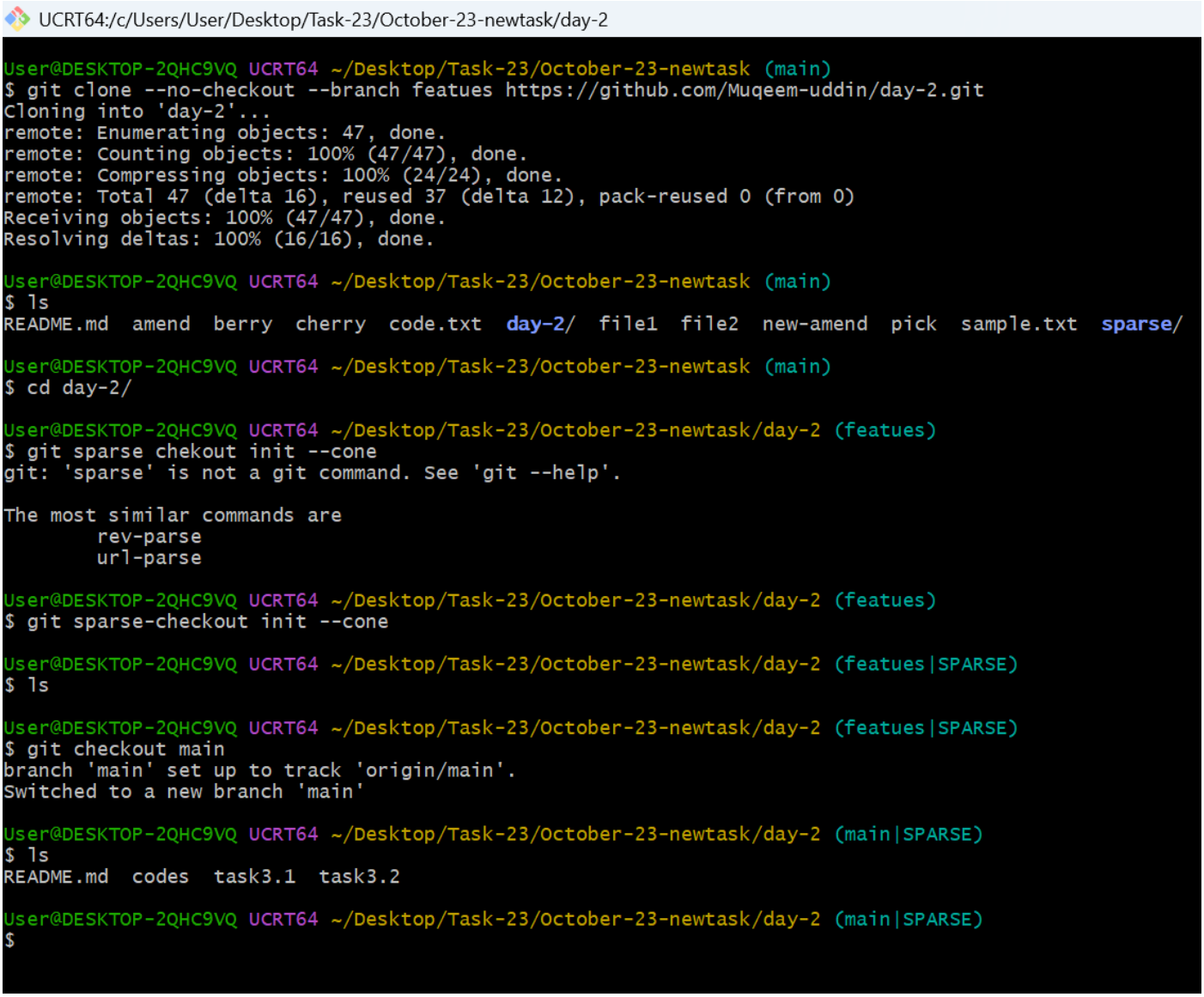
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git push origin -d v1.0
To https://github.com/viquar44/October-23-newtask.git
 - [deleted]         v1.0
```

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git tag v1.0 a6f6947

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git tag
v1.0

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git push origin v1.0
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/viquar44/October-23-newtask.git
 * [new tag]         v1.0 -> v1.0
```

8. **Clone with Sparse Checkout**
    - Clone only a subdirectory of a repo using sparse checkout.



9. **Reset vs Revert Challenge**
    - Demonstrate the difference between `git reset --hard` and `git revert` in a repo.

Git Reset : git reset is used when we want to unstage a file and bring our changes back to the working directory. git reset can also be used to remove commits from the local repository.

git reset HEAD <filename>

Whenever we unstage a file, all the changes are kept in the working area

We are back to the working directory, where our changes are present but the file is now unstaged. Now there are also some commits that we don't want to get committed and we want to remove them from our local repository. To see how to remove the commit from our local repository let's stage and commit the changes that we just did and then remove that commit.

Git Revert : git revert is used to remove the commits from the remote repository. Since now our changes are in the working directory, let's add those changes to the staging area and commit them.

Now we want to delete the commit that we just added to the remote repository. We could have used the git reset command but that would have deleted the commit just from the local repository and not the remote repository. If we do this then we would get conflict that the remote commit is not present locally. So, we do not use git reset here. The best we can use here is git revert.

git revert <commit id of the commit that needs to be removed>
Points to keep in mind -
Using git revert we can undo any commit , not like git reset where we could just remove "n" recent commits.

10. **Detached HEAD Challenge**
    - Checkout a specific commit (detached HEAD state) and create a new branch from it.

Step 1: check the logs [git log –oneline]
Step 2: use command [git

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git checkout efc3dfd
Note: switching to 'efc3dfd'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

  git switch -c <new-branch-name>

Or undo this operation with:

  git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at efc3dfd Merge pull request #5 from viquar44/nova

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask ((efc3dfd...))
$ git switch -c mars
Switched to a new branch 'mars'
```

11. **Git Hooks Challenge**
    - Configure a `pre-commit` hook to reject commits without a message format (e.g., must start with `JIRA-XXX`).

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ ls -a
./  ../  .git/  .gitignore  README.md  amend  berry  cherry  code.txt  day-2/  file1  file2  new-amend  pick  sample.txt  sparse/

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ cd .git/

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask/.git (GIT_DIR!)
$ ls
COMMIT_EDITMSG  FETCH_HEAD  HEAD  ORIG_HEAD  REBASE_HEAD  config  description  hooks/  index  info/  logs/  objects/  packed-refs  refs/

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask/.git (GIT_DIR!)
$ ls hooks/
applypatch-msg.sample*      post-update.sample*     pre-merge-commit.sample*  pre-receive.sample*         sendemail-validate.sample*
commit-msg.sample*          pre-applypatch.sample*  pre-push.sample*          prepare-commit-msg.sample*  update.sample*
fsmonitor-watchman.sample*  pre-commit.sample*      pre-rebase.sample*        push-to-checkout.sample*

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask/.git (GIT_DIR!)
$ vi commit-msg

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask/.git (GIT_DIR!)
$ chmod 755 commit-msg

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask/.git (GIT_DIR!)
$ cat commit-msg
#!/bin/sh
MESSAGE_FILE="$1"
MESSAGE=$(head -n 1 "$MESSAGE_FILE")

if ! printf '%s\n' "$MESSAGE" | grep -Eq '^JIRA-[0-9]+ '; then
    echo "ERROR: Commit message must start with JIRA-XXX"
    echo "Example: JIRA-123 Add login validation"
    exit 1
fi

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask/.git (GIT_DIR!)
$
```

12. **Squash Merge vs Rebase Merge**
    - Show the difference between squash merge and rebase merge with evidence.
    - I used ($ git merge –squash <branch name> from develop branch for whatever committed in branch master it will copy in develop branch without committing in staging are then from there can do commit all file in single commit id and single commit message.
    - Git rebase will take all commits history in linear in details it will give clean tree for commie history from that branch.

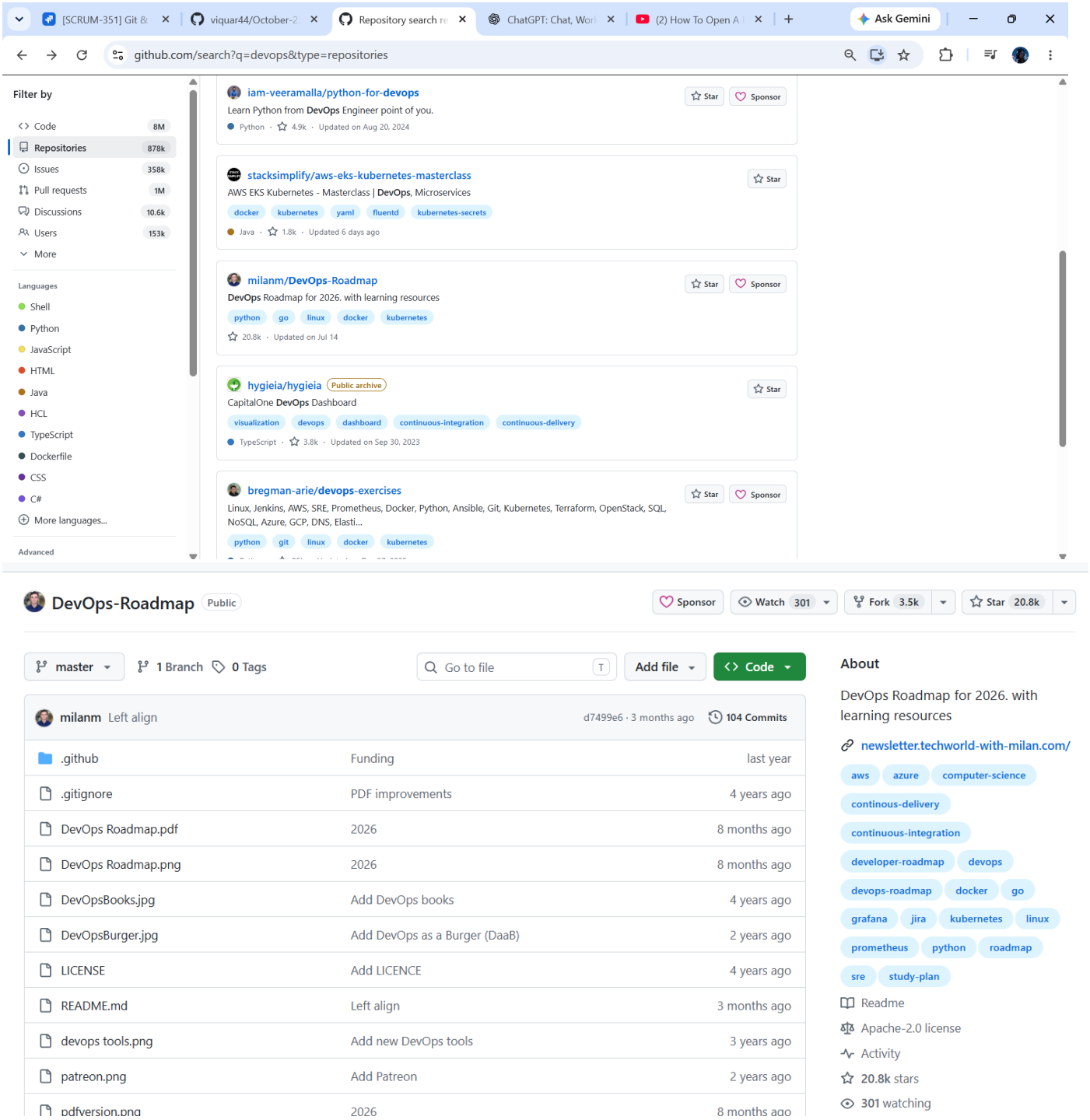
```
[root@ip-172-31-27-16 challange]# git merge --squash master
Updating 4788cde..9cb8b43
Fast-forward
Squash commit -- not updating HEAD
 1 | 0
 2 | 0
 3 | 0
 a | 0
 b | 0
 c | 0
 6 files changed, 0 insertions(+), 0 deletions(-)
 delete mode 100644 1
 delete mode 100644 2
 delete mode 100644 3
 delete mode 100644 a
 delete mode 100644 b
 delete mode 100644 c
[root@ip-172-31-27-16 challange]# git commit -m "merged from master"
[detach_branch f4d442b] merged from master
 6 files changed, 0 insertions(+), 0 deletions(-)
 delete mode 100644 1
 delete mode 100644 2
 delete mode 100644 3
 delete mode 100644 a
 delete mode 100644 b
 delete mode 100644 c
[root@ip-172-31-27-16 challange]# git log --oneline
f4d442b (HEAD → detach_branch) merged from master
4788cde (develop) added c
950c011 added b
d9a3585 added a
c465bdc commited using squash
9e034c0 deleted
74f7287 commit from develop
dede469 detach_practise1
1310a6e detach_practise
ca076ab Revert "revert_practise"
488030e revert
e550fda (tag: v1.0, tag: show) ggg
d71449c added 1,2,3 blank files
20d5728 commited from master
fd694eb comited
```

```
[root@ip-172-31-27-16 challange]# git log --oneline
71f871e (HEAD → develop) JIRA-123 added devops
15b40bd JIRA-123 added horizone
3d66202 JIRA-123 added techie
861bdd0 JIRA-123 commiting
148a6da JIRA-123 added
4788cde added c
950c011 added b
d9a3585 added a
c465bdc commited using squash
9e034c0 deleted
```

```
Switched to branch 'detach_branch'
[root@ip-172-31-27-16 challange]# git rebase develop
Successfully rebased and updated refs/heads/detach_branch.
[root@ip-172-31-27-16 challange]# git status
On branch detach_branch
nothing to commit, working tree clean
[root@ip-172-31-27-16 challange]# git log --oneline
88bb418 (HEAD → detach_branch) added z
9906f1d added y
1d039bb added x
5c24b82 merged from master
71f871e (develop) JIRA-123 added devops
15b40bd JIRA-123 added horizone
3d66202 JIRA-123 added techie
861bdd0 JIRA-123 commiting
148a6da JIRA-123 added
4788cde added c
950c011 added b
d9a3585 added a
c465bdc commited using squash
9e034c0 deleted
74f7287 commit from develop
dede469 detach_practise1
1310a6e detach_practise
ca076ab Revert "revert_practise"
488030e revert
e550fda (tag: v1.0, tag: show) ggg
d71449c added 1,2,3 blank files
20d5728 commited from master
fd694eb comited
4a88e46 commited from master
df302bd commited
69d77fb this is commited from master branch
bea8bc3 this is commited from develop branch
```

13. Fork & Pull Request Workflow

- Fork a repo, make a change, and submit a pull request to the original repo.



Create a new fork

A *fork* is a copy of a repository. Forking a repository allows you to freely experiment with changes without affecting the original project. View existing forks.

Required fields are marked with an asterisk (*).

Owner * viquar44 / **Repository name *** DevOps-Roadmap

DevOps-Roadmap is available.

By default, forks are named the same as their upstream repository. You can customize the name to distinguish it further.

Description

DevOps Roadmap for 2026. with learning resources

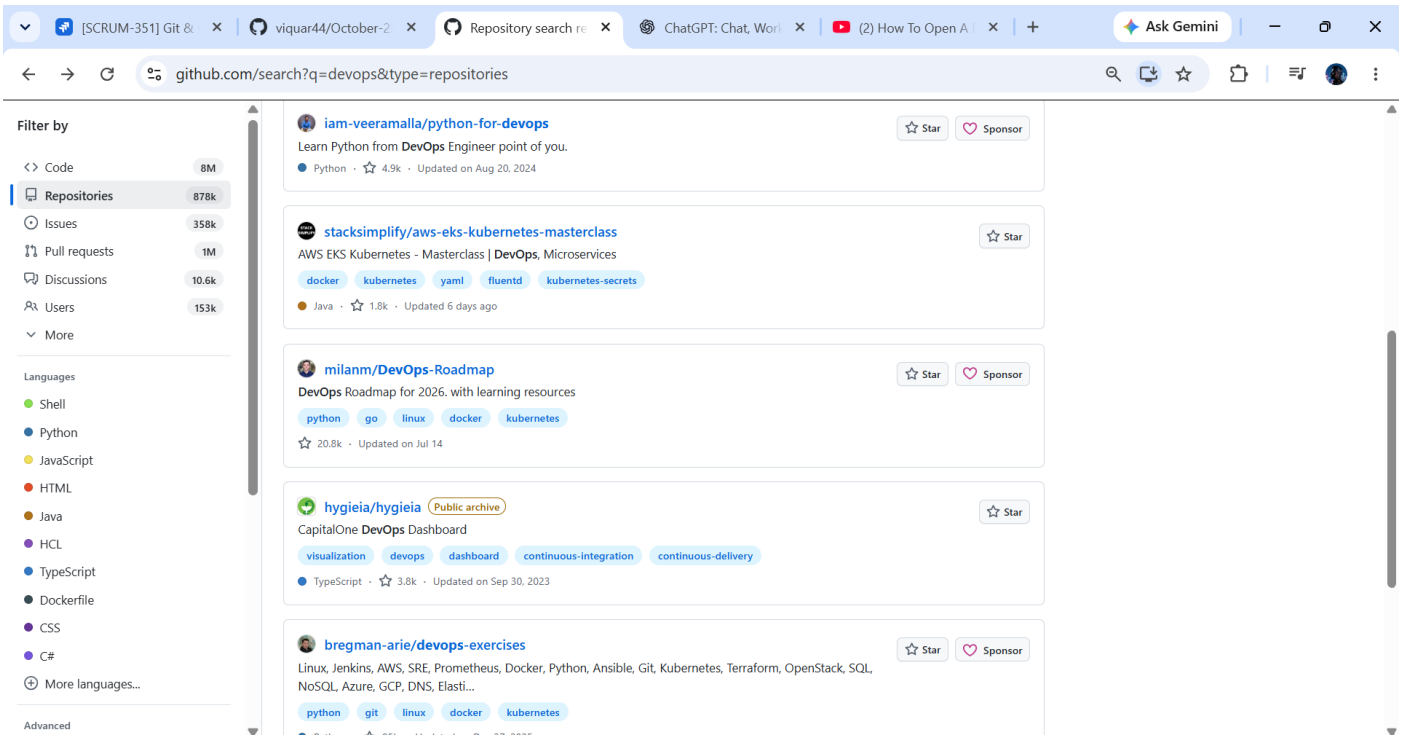
48 / 350 characters

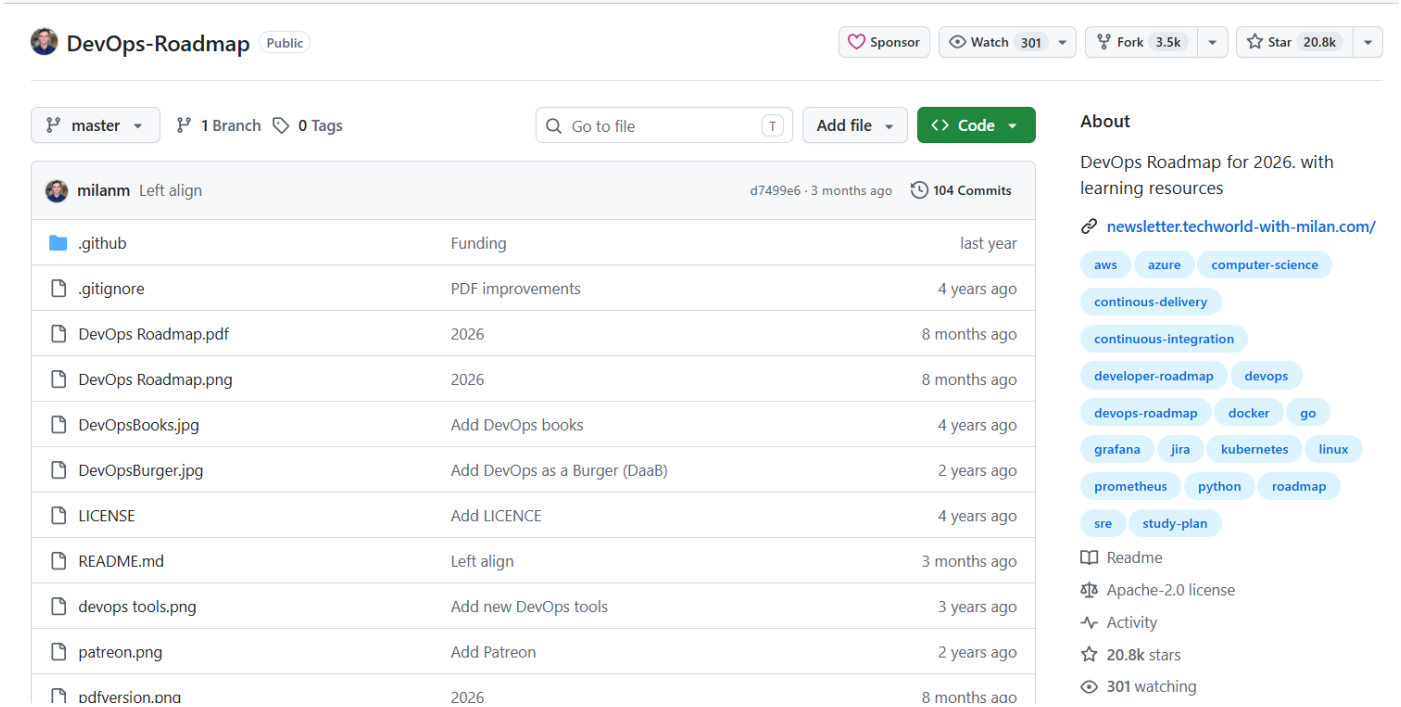
☑ Copy the `master` branch only

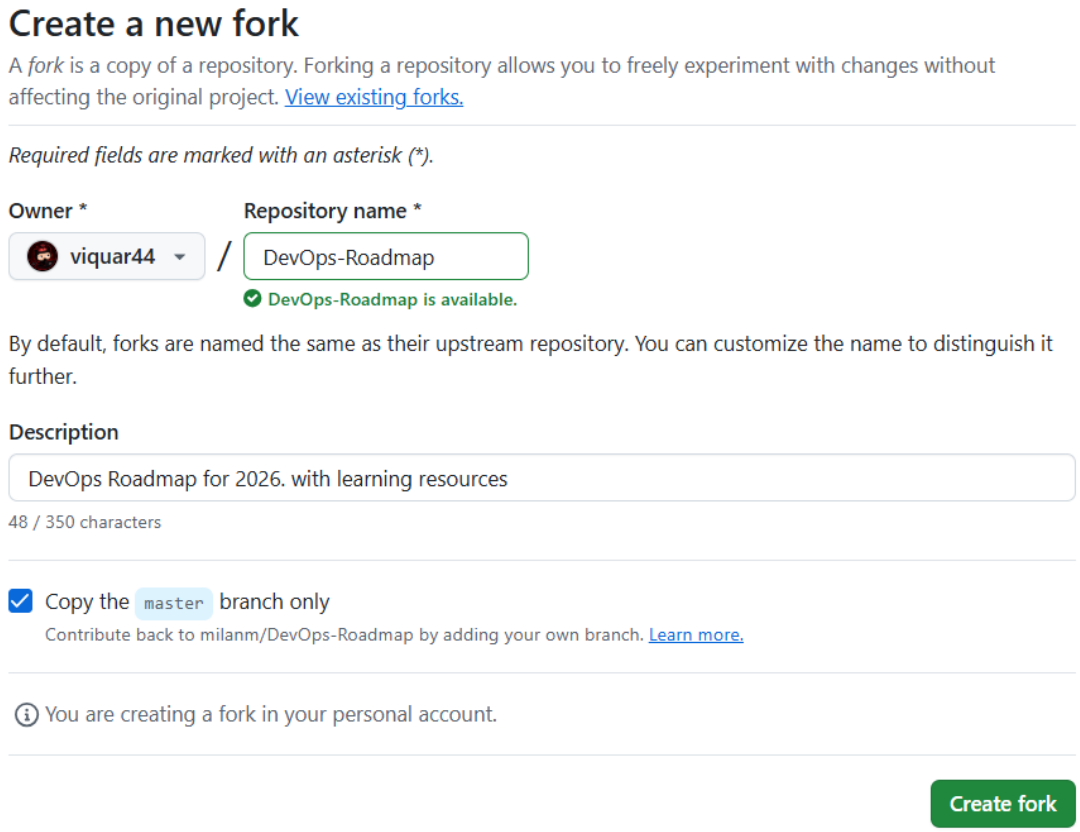
Contribute back to milanm/DevOps-Roadmap by adding your own branch. Learn more.

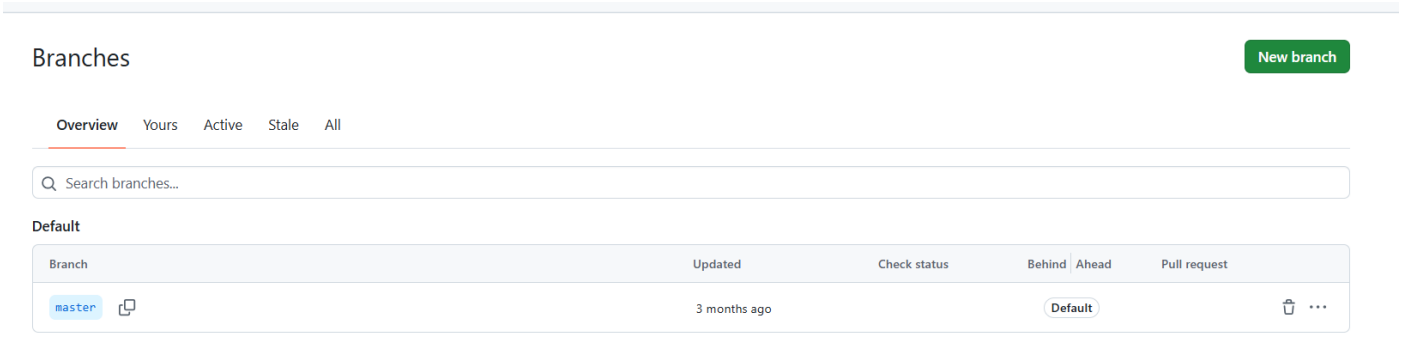
You are creating a fork in your personal account.

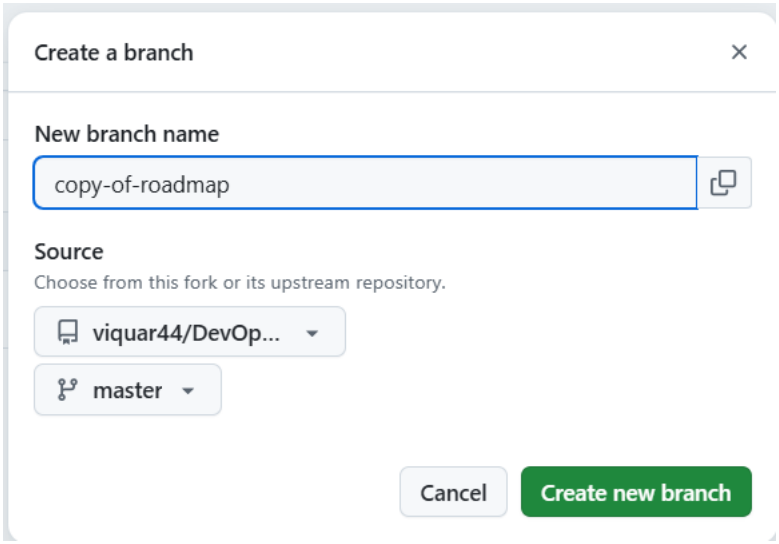
Create fork

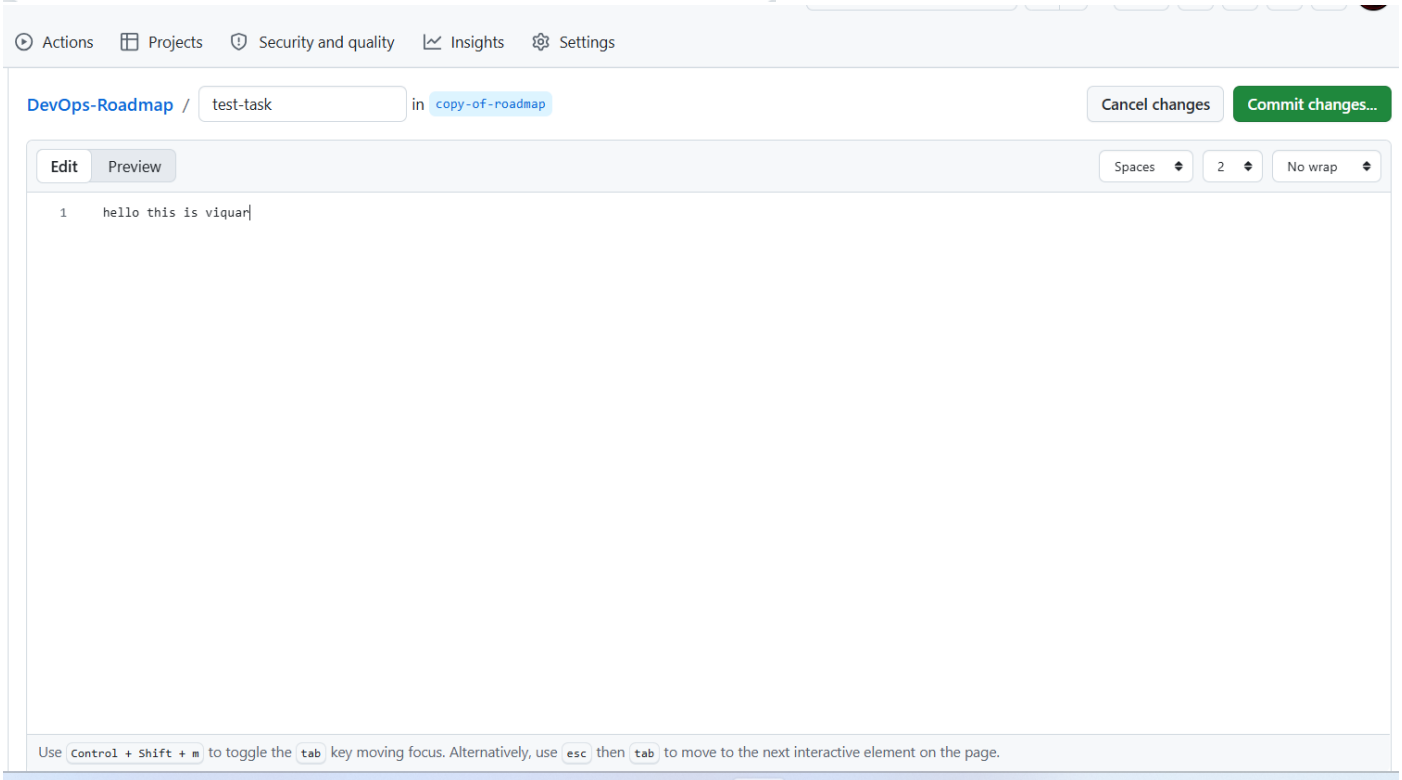


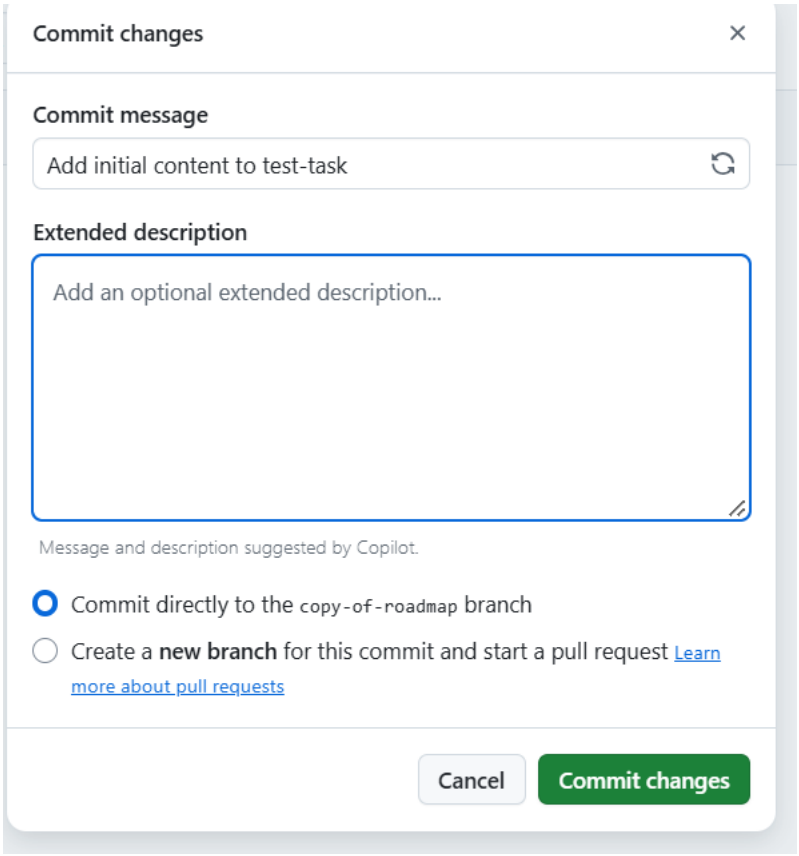

DevOps-Roadmap
Public
Sponsor
Watch 301
Fork 3.5k
Star 20.8k
master
1 Branch
0 Tags
Go to file
Add file
Code
milanm Left align
d7499e6 · 3 months ago
104 Commits
.github Funding last year
.gitignore PDF improvements 4 years ago
DevOps Roadmap.pdf 2026 8 months ago
DevOps Roadmap.png 2026 8 months ago
DevOpsBooks.jpg Add DevOps books 4 years ago
DevOpsBurger.jpg Add DevOps as a Burger (DaaB) 2 years ago
LICENSE Add LICENCE 4 years ago
README.md Left align 3 months ago
devops tools.png Add new DevOps tools 3 years ago
patreon.png Add Patreon 2 years ago
pdfversion.png 2026 8 months ago
About
DevOps Roadmap for 2026. with learning resources
newsletter.techworld-with-milan.com/
aws
azure
computer-science
continous-delivery
continuous-integration
developer-roadmap
devops
devops-roadmap
docker
go
grafana
jira
kubernetes
linux
prometheus
python
roadmap
sre
study-plan
Readme
Apache-2.0 license
Activity
20.8k stars
301 watching

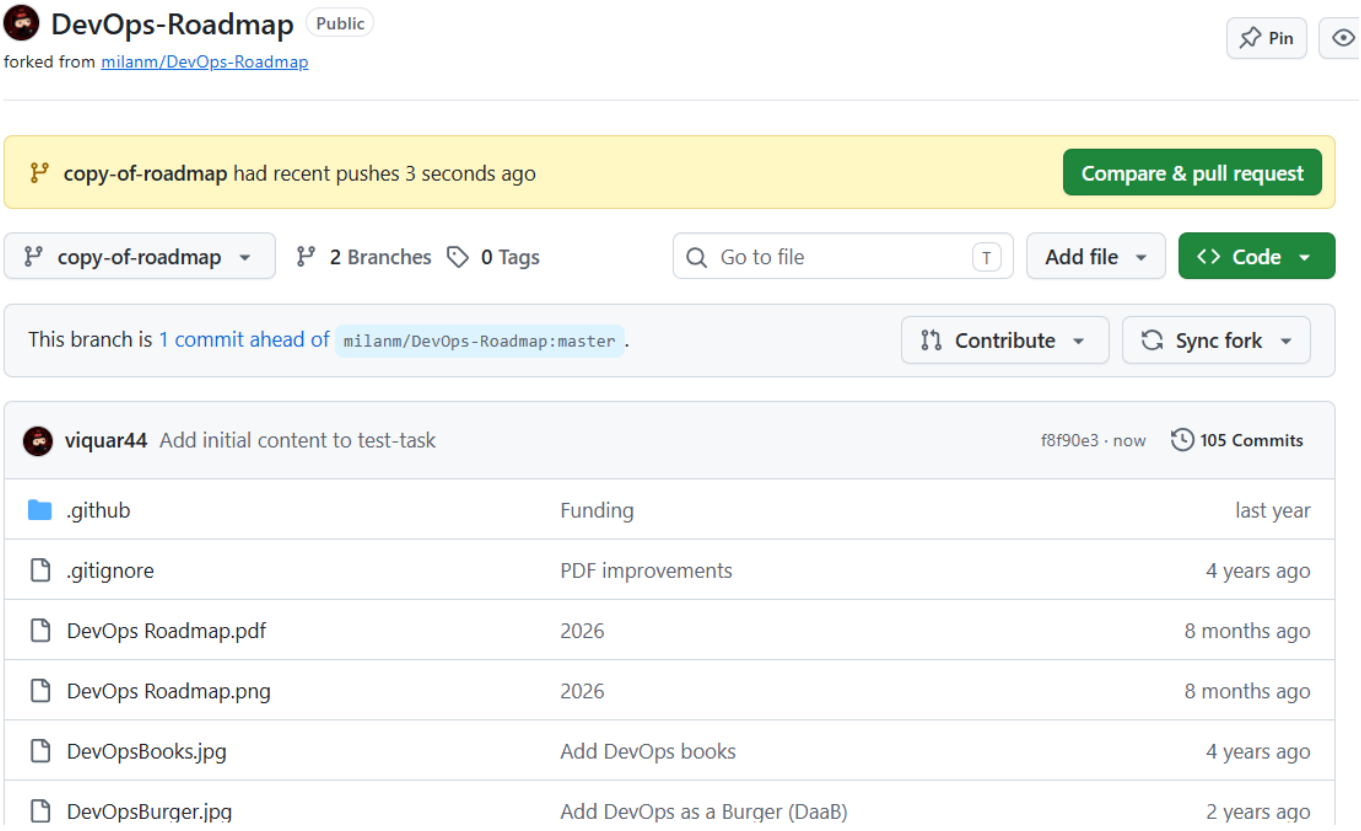

Create a new fork
A fork is a copy of a repository. Forking a repository allows you to freely experiment with changes without affecting the original project. View existing forks.
Required fields are marked with an asterisk (*).
Owner *
Repository name *
viquar44
/
DevOps-Roadmap
DevOps-Roadmap is available.
By default, forks are named the same as their upstream repository. You can customize the name to distinguish it further.
Description
DevOps Roadmap for 2026. with learning resources
48 / 350 characters
Copy the master branch only
Contribute back to milanm/DevOps-Roadmap by adding your own branch. Learn more.
You are creating a fork in your personal account.
Create fork

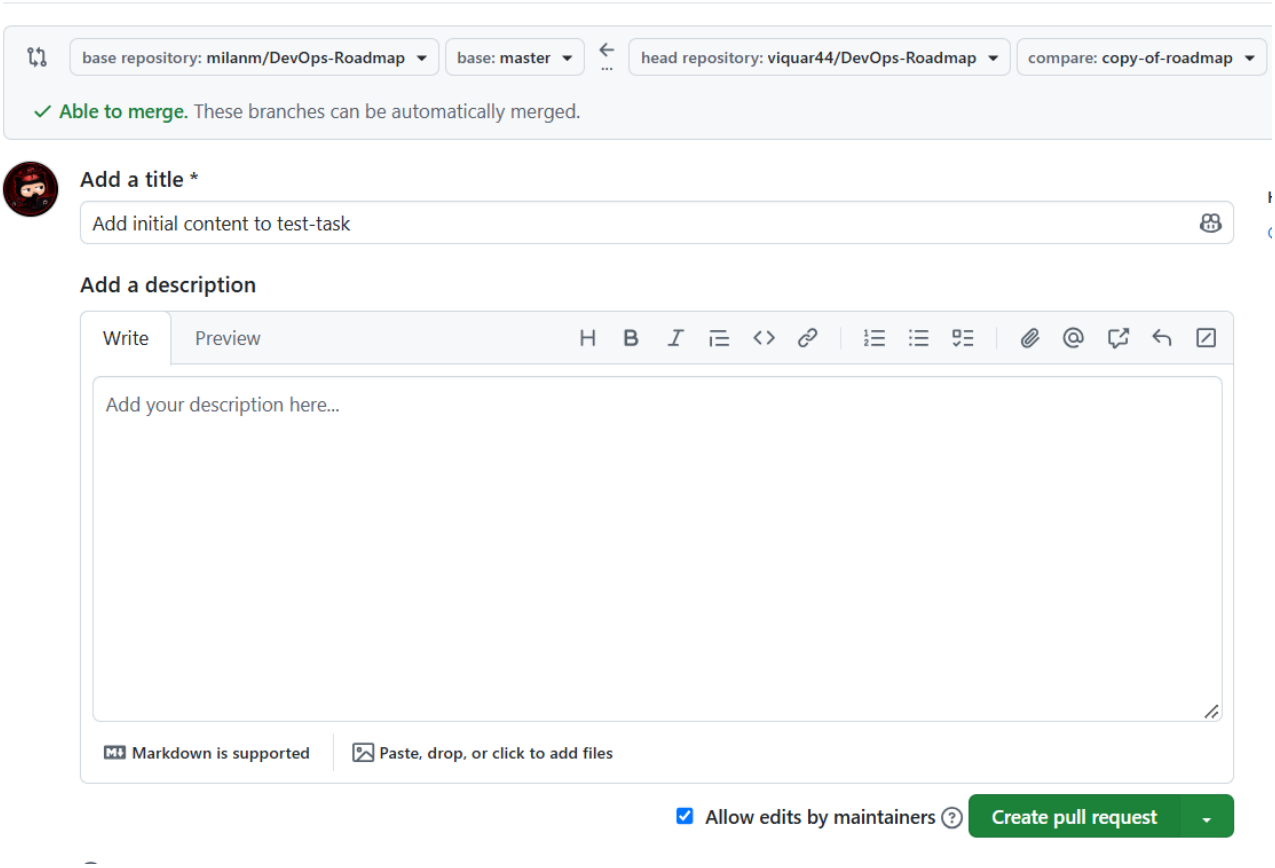

Branches
New branch
Overview
Yours
Active
Stale
All
Search branches...
Default
Branch
Updated
Check status
Behind
Ahead
Pull request
master
3 months ago
Default

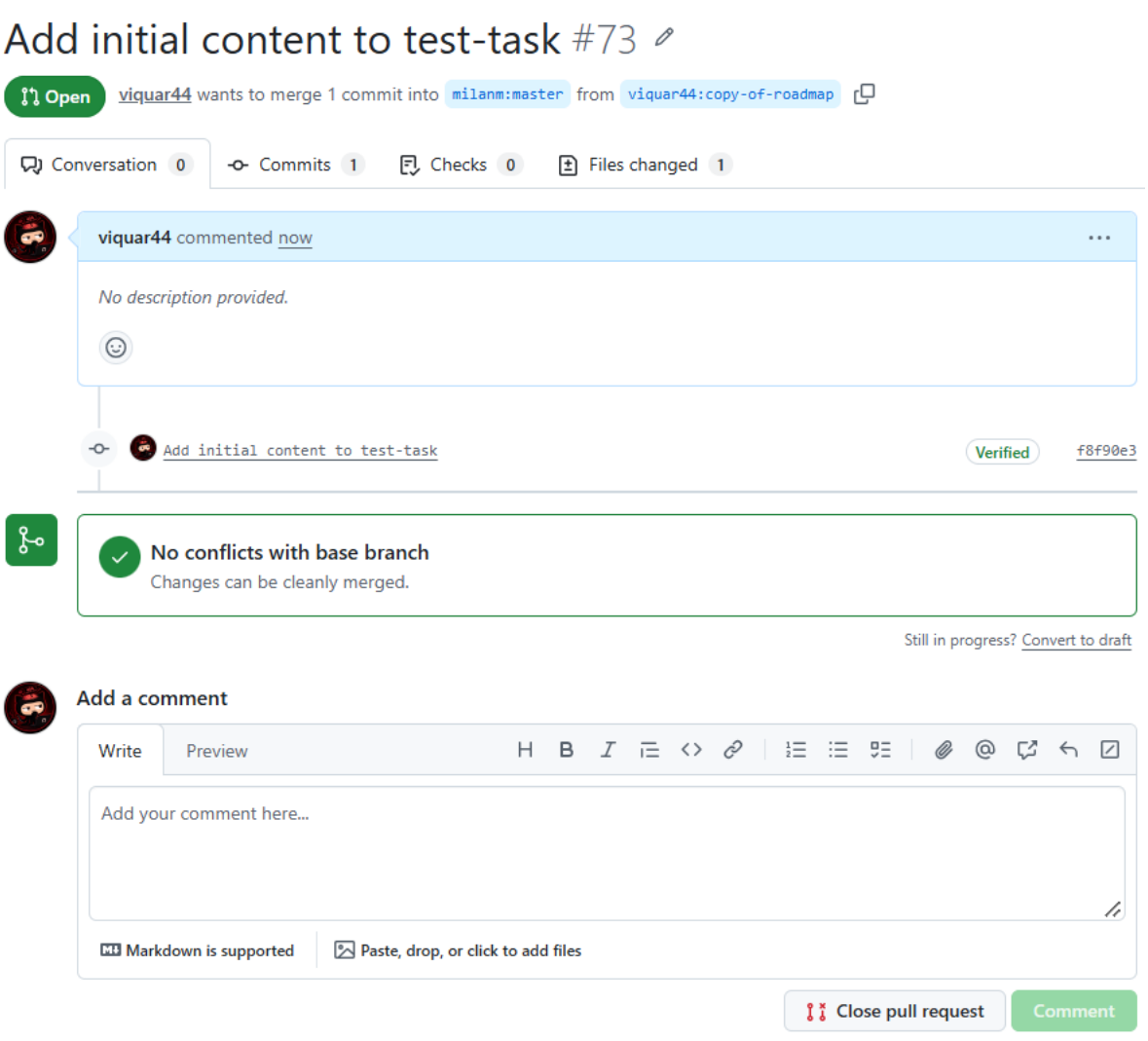

Create a branch
New branch name
copy-of-roadmap
Source
Choose from this fork or its upstream repository.
viquar44/DevOp...
master
Cancel
Create new branch


Actions
Projects
Security and quality
Insights
Settings
DevOps-Roadmap
/
test-task
in
copy-of-roadmap
Cancel changes
Commit changes...
Edit
Preview
Spaces
2
No wrap
1
hello this is viquar
Use Control + Shift + m to toggle the tab key moving focus. Alternatively, use esc then tab to move to the next interactive element on the page.


Commit changes
Commit message
Add initial content to test-task
Extended description
Add an optional extended description...
Message and description suggested by Copilot.
Commit directly to the copy-of-roadmap branch
Create a new branch for this commit and start a pull request Learn more about pull requests
Cancel
Commit changes


DevOps-Roadmap
Public
forked from milanm/DevOps-Roadmap
Pin
copy-of-roadmap had recent pushes 3 seconds ago
Compare & pull request
copy-of-roadmap
2 Branches
0 Tags
Go to file
T
Add file
Code
This branch is 1 commit ahead of milanm/DevOps-Roadmap:master.
Contribute
Sync fork
viquar44 Add initial content to test-task
f8f90e3 · now
105 Commits
.github Funding last year
.gitignore PDF improvements 4 years ago
DevOps Roadmap.pdf 2026 8 months ago
DevOps Roadmap.png 2026 8 months ago
DevOpsBooks.jpg Add DevOps books 4 years ago
DevOpsBurger.jpg Add DevOps as a Burger (DaaB) 2 years ago


base repository: milanm/DevOps-Roadmap
base: master
head repository: viquar44/DevOps-Roadmap
compare: copy-of-roadmap
Able to merge. These branches can be automatically merged.
Add a title *
Add initial content to test-task
Add a description
Write
Preview
Add your description here...
Markdown is supported
Paste, drop, or click to add files
Allow edits by maintainers
Create pull request


Add initial content to test-task #73
Open
viquar44 wants to merge 1 commit into milanm:master from viquar44:copy-of-roadmap
Conversation 0
Commits 1
Checks 0
Files changed 1
viquar44 commented now
No description provided.
Add initial content to test-task
Verified
f8f90e3
No conflicts with base branch
Changes can be cleanly merged.
Still in progress? Convert to draft
Add a comment
Write
Preview
Add your comment here...
Markdown is supported
Paste, drop, or click to add files
Close pull request
Comment

14. **Recover Lost Commit**
    - Commit something, reset hard, and then recover it using `git reflog`.

```
        new file:   reset


User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git commit -m "new files to be add"
[main fd61df6] new files to be add
 2 files changed, 1 insertion(+)
 create mode 160000 day-2
 create mode 100644 reset

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git log --oneline
fd61df6 (HEAD -> main) new files to be add
b8bf43e (origin/main, origin/HEAD) add sparse me
a6f6947 (tag: v1.0) Merge branch 'main' of https://github.com/viquar44/October-23-newtask
46f489b berry , cherry & pick added to commit by main
efc3dfd (mars) Merge pull request #5 from viquar44/nova
```

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git reset --hard b8bf43e
warning: unable to rmdir 'day-2': Directory not empty
HEAD is now at b8bf43e add sparse me

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git log --oneline
b8bf43e (HEAD -> main, origin/main, origin/HEAD) add sparse me
a6f6947 (tag: v1.0) Merge branch 'main' of https://github.com/viquar44/October-23-newtask
46f489b berry , cherry & pick added to commit by main
efc3dfd (mars) Merge pull request #5 from viquar44/nova
7eedc69 Merge pull request #6 from viquar44/mercury
403a954 pick added to commit in mercury
```

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ ls
README.md  amend  berry  cherry  code.txt  day-2/  file1  file2  new-amend  pick  sample.txt  sparse/

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git reflog
b8bf43e (HEAD -> main, origin/main, origin/HEAD) HEAD@{0}: reset: moving to b8bf43e
fd61df6 HEAD@{1}: commit: new files to be add
b8bf43e (HEAD -> main, origin/main, origin/HEAD) HEAD@{2}: checkout: moving from mars to main
efc3dfd (mars) HEAD@{3}: checkout: moving from efc3dfd457f8018cca95f85f90fd5d8e24b3e50e to mars
efc3dfd (mars) HEAD@{4}: checkout: moving from main to efc3dfd
b8bf43e (HEAD -> main, origin/main, origin/HEAD) HEAD@{5}: commit: add sparse me
a6f6947 (tag: v1.0) HEAD@{6}: pull: Merge made by the 'ort' strategy.
46f489b HEAD@{7}: commit (amend): berry , cherry & pick added to commit by main
```

```
User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ git reset --hard fd61df6
HEAD is now at fd61df6 new files to be add

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$ ls
README.md  amend  berry  cherry  code.txt  day-2/  file1  file2  new-amend  pick  reset  sample.txt  sparse/

User@DESKTOP-2QHC9VQ UCRT64 ~/Desktop/Task-23/October-23-newtask (main)
$
```